Extensive canonical-sequence overlap and unresolved source attribution in a public non-canonical HLA-peptide atlas

Rom Jan

Every number is regenerated from the analysis artifacts by `verify_manuscript.py`, which fails the build on drift. Every quotation from a primary Methods section has been verified in the fetched full text.

> ## What was known / What this study adds
>
> The underlying principles here are established, and we claim none of them.
>
> **Known.** A peptide matching a canonical protein does not identify a non-canonical source and should be **excluded** before being called non-canonical (**Bedran et al. 2023**; **Aggarwal et al. 2022** — full quotations in the Introduction); the underlying protein-inference problem is textbook (**Nesvizhskii & Aebersold 2005**); pooled target–decoy competition can distribute errors unevenly across classes, requiring group-aware assessment and control (**Woo et al. 2014**; **Group-walk**, Freestone et al. 2022; Zhang & Bassani-Sternberg 2023; **pXg**, Choi & Paek 2024).
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> **Our contribution is empirical:** an audit of whether a major public resource satisfies these principles, a measurement of the sequence ambiguity latent in its peptide catalogue and in one of its source libraries, a within-resource test of the consequence, and an atlas-level implementation of existing MIAIPE/mzIdentML provenance fields. Aggarwal et al. state that the problem is *not quantified* and that *no specific databases are criticised by name*; Bedran et al., who did quantify, did not include IEAtlas.

Abstract

Catalogues of non-canonical (ncORF-derived) HLA-presented peptides are used to nominate cancer-vaccine targets. A peptide sequence also encoded by a canonical human protein does not uniquely identify a non-canonical source, since tandem MS identifies a *sequence*, not a locus. The field's own published, recommended remedy is to **exclude** such sequences before calling a peptide non-canonical (Bedran et al. 2023; Aggarwal et al. 2022); our contribution is testing whether a major public resource satisfies it.

We audit **IEAtlas**, not previously examined against this criterion. **98,193 of 174,465 unique cancer-catalogued peptide sequences (56.3%)** exactly match a protein in a frozen reviewed human reference *R*, and did so within IEAtlas's own search database — its Methods describe searching spectra against both its ncORF library and the canonical human proteome — an internal finding, not a comparison against a reference it never consulted. These matches do not establish canonical production or that any resource acted improperly; they show the sequence cannot be uniquely attributed to its nominated source from sequence and MS evidence alone.

Latent canonical ambiguity varies enormously across ncORF libraries — **34.1%** of nuORFdb v1.2's distinct 9-mers against **1.0–2.4%** for GENCODE Ribo-seq ORFs — but does not determine a catalogue's rate: the exclusion step does.

The consequence is observable inside the resource: canonical-compatible sequences appear in IEAtlas's **own** normal-tissue export at **22.4%** versus **9.1%** for canonical-incompatible ones; after joint standardization by peptide length and cancer-detection breadth, the risk ratio is **1.72×** (gene-clustered bootstrap **95% CI [1.66, 1.80]**). **22,003 sequences (12.6% of the catalogue)** are both canonical-compatible and already in that export — consistent with, but not specific to, greater detectability, and warranting normal-presentation review before treatment as a tumour-restricted target.

Separately and additively, IEAtlas's pooled 5% PSM-level FDR cannot be resolved to a class-specific estimate for its non-canonical subset without class-resolved target–decoy counts, which it does not publish. We propose retaining shared sequences with an explicit source-compatibility annotation instead of treating them as uniquely non-canonical, and losslessly propagating established MIAIPE/mzIdentML provenance into atlas exports. In a separate raw-linked ImmunoVerse pilot, two recurrent peptide sequences had current accepted scan evidence in every catalogue-listed cell-line context, yet those contexts shared no reported HLA allele; sequence recurrence therefore did not represent recurrence of one peptide–HLA allomorph. Together, these results show that spectrum-assignment confidence, source attribution and target recurrence are distinct estimands: confidence at one level does not automatically survive atlas aggregation.

Introduction

Non-canonical open reading frames (ncORFs) — in long non-coding RNAs, pseudogenes, untranslated regions, and alternative frames of coding genes — yield peptides that are presented on HLA molecules and recognised by T cells. Because their products are absent from the annotated proteome, they have been proposed as an unusually attractive class of tumour antigen: potentially tumour-restricted, shared across patients, and not subject to central tolerance. Several public catalogues now aggregate tens or hundreds of thousands of such epitopes, and these catalogues are the practical input to target selection.

Building such a catalogue requires answering one question for every identified peptide: **is this peptide non-canonical?** The question is harder than it appears, and the field has known why for two decades.

Tandem MS identifies a sequence, not a source. A peptide-spectrum match establishes an amino-acid sequence. It does not establish which genomic locus produced it. Where two loci encode the same sequence — a canonical protein and an ncORF — no spectral evidence distinguishes them. This is the protein-inference problem (Nesvizhskii & Aebersold), and in proteogenomics it is *"exacerbated by the mapping complexity where many identified peptides map to several loci, both novel and known"* (Aggarwal et al. 2022).

A criterion is published and recommended. Aggarwal et al. (2022) state that *"most shared peptides should be dropped if defining… novel-coding regions."* Bedran et al. (2023) implement the rule directly: sequences perfectly matching any known protein are treated as *exonic* and excluded, with a stringent requirement of *"at least three mismatches with any known protein sequence"* before a peptide is called non-canonical. A 2026 TransCODE consortium paper implements the same principle for HLA immunopeptidomics specifically: a peptide is classified as ncORF-derived only if it maps (≥8 aa; ≤30 distinct Swiss-Prot mappings for canonical, ≤10 distinct ncORF mappings for ncORF-derived) to a ncORF and not a canonical protein — its own criteria, which the paper itself describes as *"less strict than those used by PeptideAtlas to avoid mapping canonical protein derived peptides to ncORFs"* (Deutsch et al. 2026): PeptideAtlas's own production scheme is stricter still, and exists for exactly this purpose. BamQuery (Ruiz Cuevas et al. 2023) applies a related default at the tool level rather than the atlas level: its best-guess biotype calls a peptide canonical whenever at least one in-frame canonical region can explain it, a rule its own authors show can override likelihood evidence pointing the other way for a real fraction of the peptides it calls canonical. The underlying rationale presupposes the search process does not do this on its own, exactly as Bedran's own explicit mismatch threshold does. We call this a **sequence-exclusivity criterion**. It is published and recommended; we do not assert it is a universal rule binding every atlas. An atlas may legitimately retain shared observations — provided it labels them as source-ambiguous rather than as sequence-unique ncORF products. That distinction is the subject of this paper.

The reporting remedy is also prior art. MIAIPE already recommends documenting the searched database and decoy strategy, stating whether peptide mappings are unique, reporting every possible source protein for ambiguous mappings, and depositing processed identification results (Lill et al. 2018). mzIdentML can represent a peptide's multiple database-sequence mappings, decoy status and protein ambiguity groups (Vizcaíno et al. 2017), but this capacity goes unused by default: every standard Philosopher report writer (PSM, ion, peptide, protein, MetaMSstats) excludes decoy rows entirely unless decoy-inclusion is explicitly requested (`lib/rep/psm.go`; the CLI flag itself defaults to false, `cmd/report.go`) — so the per-class decoy count exists nowhere but a discarded run-log line, at any stage of one widely-used pipeline's own standard output, by default. Our standards contribution is therefore not a new minimum-information vocabulary. It is to quantify what is lost when established search evidence is collapsed during atlas aggregation, and to specify an atlas-level provenance-retention profile.

A parallel and additive standard governs statistical confidence: a pooled target–decoy threshold under-controls a minority class, so class-specific FDR estimation is required (Woo et al. 2014, who measured 36% novel-class FDR at a 1% combined threshold; Zhang & Bassani-Sternberg 2023; Choi & Paek 2024). The same mechanism recurs outside immunopeptidomics: in bulk proteome-wide noncanonical-protein detection, Wacholder & Carvunis (2023) show, in *S. cerevisiae* data, that a pooled 1% FDR threshold can correspond to roughly 56% FDR within the noncanonical subset alone, applying the same canonical-exclusion principle to a different search context. We cite this yeast estimate for the mechanism it demonstrates — pooled thresholds under-controlling a minority class — not as a human or immunopeptidomics-specific figure.

What has not been done is to check whether the public resources satisfy these criteria. Aggarwal et al. review the problem without quantifying it, and note that no specific databases are criticised by name. Bedran et al. did quantify — reporting residual canonical overlap of 1.4% (Erhard et al. 2018), 3% (Ouspenskaia et al. 2022), 4% (Chong et al. 2020) and 5% (Laumont et al. 2016) — but their comparison did not include IEAtlas, the largest such atlas and the one most readily used as an off-the-shelf source of candidate antigens.

This paper is that check.

Results

R1. Most of IEAtlas's cancer-catalogued sequences also occur in canonical proteins

The measurement. A catalogued peptide counts as canonical-compatible if it is an **exact substring** of at least one protein in a frozen reviewed human reference R (release and hash in Methods). The complementary set — catalogued sequences that are not an exact substring of any protein in R — are termed **canonical-incompatible** in what follows: a reference-relative label, not a proven absence of a canonical source, since a broader reference could reclassify some of them as compatible (Discussion). Peptides are deduplicated to **unique sequences**; the unit is a sequence, not an atlas row.

98,193 of 174,465 unique cancer-catalogued sequences (56.3%) are canonical-compatible (**Figure 1**).

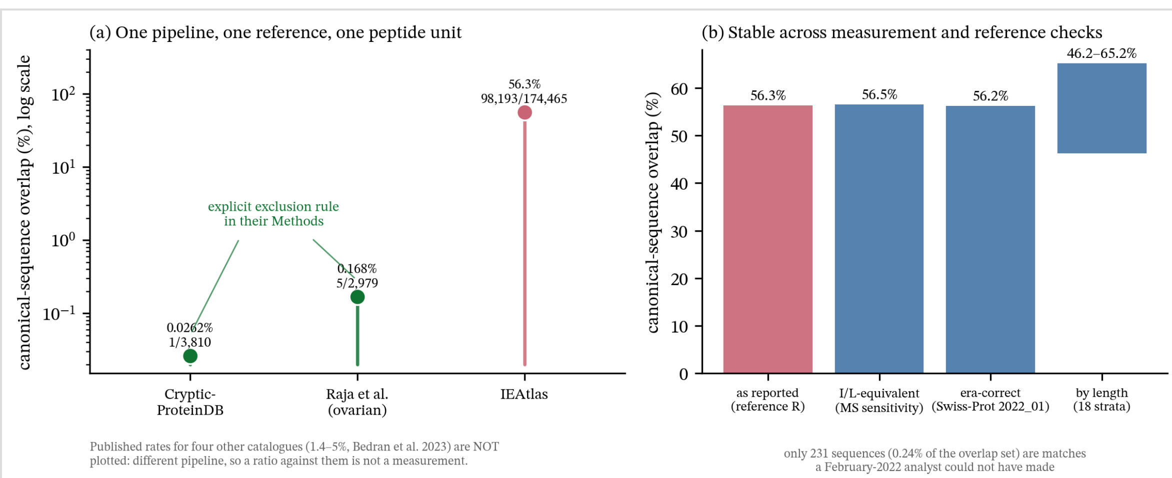

Figure 1. Left: the canonical-overlap rate under one pipeline, one reference and one peptide unit, against two catalogues that apply an explicit exclusion rule versus IEAtlas, which does not describe one. Right: I/L-equivalent matching, the era-correct reference and per-length strata; the overlap is stable across these measurement and reference checks.

Four checks establish that this is not an artifact of how we measured it.

robustness check	result
headline, reference *R* (current reviewed human proteome)	**56.3%** (98,193 / 174,465)
I/L-equivalent (nested MS-equivalence sensitivity)	**56.5%** (98,546 / 174,465; +353 sequences)
era-correct: Swiss-Prot **2022_01**, a plausible candidate for the release IEAtlas searched	**56.2%** (97,999 / 174,465)
by length (18 strata, 8–25 aa, each $n \geq 30$)	**46.2%–65.2%** — high throughout, not one band
class composition: if the atlas contained no pseudogene ORFs at all	**55.8%**

Table 1. Four robustness checks against the headline 56.3% overlap rate: none moves it materially.

The I/L row is a measurement-equivalence sensitivity analysis, not a redefinition of the primary estimand. Because routine tandem MS generally cannot distinguish leucine from isoleucine, both residues were replaced by one symbol before matching, following the sensitivity convention used by Bedran et al. (2023). This adds only **353 sequences (0.2 percentage points)**. We retain the literal exact-string result as the headline because it is simplest and directly auditable; if anything, it is slightly conservative with respect to routine MS distinguishability.

The era check settles the anachronism question empirically. Sequence novelty is reference-relative, so an overlap we score today might reflect a canonical protein that entered the reference *after* IEAtlas was built — in which case faulting the atlas would be anachronistic. It does not. Rebuilding the reference at Swiss-Prot 2022_01 (20,376 human proteins) moves the rate by **0.1 percentage points**; only **231 sequences (0.24% of the overlap set)** are matches that a 2022 analyst could not have made. The overlap was almost entirely visible at build time.

The 231 and the 0.1 pp both reconcile, and we show the arithmetic rather than leave the reader to take the net shift on faith. Going from Swiss-Prot 2022_01 to the modern release is not pure growth: **231 sequences** gained a canonical match (present only in the modern reference) and **37** *lost* one (matched a 2022_01 entry later retired, merged, or resequenced) — a **net** change of 231 − 37 = **194** sequences, exactly the difference between the two headline counts (98,193 − 97,999 = 194). We report the net rate shift (+0.1 pp) as the headline because that is what answers the anachronism question; the gross 231/37 is reported here because "net" and "gross" are not interchangeable and a reader checking our arithmetic should not have to guess which one a given number means.

The canonical proteins were in IEAtlas's own search database

The era check above understates the point, because it treats Swiss-Prot 2022_01 as a *reference we chose*. It is not: whichever exact Swiss-Prot release IEAtlas downloaded, it is **the canonical half of IEAtlas's own search database.** From its Methods:

> The RAW MS data files were downloaded and analyzed by MaxQuant (v.2.1.0.0) [...] Files were searched against **both** our integrated benchmarked ncORF library **and the canonical human proteome** obtained from the UniProt database with Swiss-Prot protein evidence (downloaded in February 2022). [...] Only epitopes derived from non-coding regions were retained.

Which release "downloaded in February 2022" identifies is genuinely ambiguous, and we do not overclaim it. UniProt's release immediately before 2022_01 was **2021_04** (current 17 Nov 2021 – 22 Feb 2022); **2022_01** became current only on **23 Feb 2022**. So of February's 28 days, 22 fall inside 2021_04's window and only 6 inside 2022_01's — if anything, the calendar mildly favors 2021_04, the release named for November, over 2022_01, the release named for February, which is the reverse of what naming alone suggests. We built our era-correct comparator at 2022_01 and label it a **plausible candidate**, not an identified match; we have not built 2021_04 and do not assert its overlap rate. What we do assert: total Swiss-Prot grew by only 1,068 entries (≈0.19%) across the entire 14-week gap between these two releases, worldwide and all organisms combined — smaller than the movement between 2022_01 and the *modern* release measured directly above (which spans several years and materially more growth, and still only shifted the overlap rate by 0.1 pp). Whichever of the two February-2022 candidates IEAtlas actually used, the era-correct check's own finding — that Swiss-Prot vintage barely moves this rate — makes it implausible that the choice between them would move it by anything approaching 0.1 pp either. We report the 2022_01 build as the concrete comparator because a comparator has to be built at some point, not because we can name the exact release with confidence.

MaxQuant is named directly in that Methods passage (quoted in full above, not assumed). Given several FASTA inputs, it concatenates them into a single target database; every spectrum is scored against canonical and non-canonical candidates **together**. So for each of the **97,999 (56.2%)** catalogued sequences that occur in a canonical human protein — in our plausible-candidate reconstruction of the release, not a confirmed identical copy of IEAtlas's own file — an entry carrying that identical sequence was physically present in the database the spectrum was matched against, alongside the ncORF to which the peptide was ultimately attributed.

Nothing here depends on that being one search rather than two. Were the two FASTAs instead searched separately and compared, the canonical proteome would still have been consulted by the pipeline that produced the catalogue, and the overlap would still have been visible to it. The claim is only that the canonical sequences were inside the procedure, not outside it — which is what the Methods say.

What we reconstructed, and what we did not obtain. We do not hold IEAtlas's literal search FASTA — the exact byte-for-byte file its MaxQuant run consumed, with its specific isoform-inclusion settings, accession list and header format. What we hold is a **release-matched reconstruction**: Swiss-Prot 2022_01 human entries (OX=9606), rebuilt live from UniProt's own archived release, matching what the Methods describe downloading. A rebuilt release and the literal searched file are not guaranteed identical — UniProt's isoform-inclusion defaults, an unrecorded canonical-only-vs-all-isoforms toggle, or a differently scoped taxonomy query could each produce a slightly different entry set from the same nominal release. We do not have IEAtlas's `peptides.txt` or database hash to check this directly. The era-correct check above is therefore the load-bearing evidence for this claim, not the modern reference: it shows the reconstructed 2022_01 database and the modern one agree to 0.1 pp, so the finding is not sensitive to exactly which entries a given Swiss-Prot build snapshot happens to carry. We consider this sufficient to say the canonical proteins were present in *a* database matching what IEAtlas describes searching, and we phrase the claim that way rather than asserting identity with a file we have never seen. Obtaining IEAtlas's own `peptides.txt` would settle this outright (§ Discussion, "Evidence required for direct verification").

This changes what kind of claim the paper is making. The source ambiguity is not something an external auditor reconstructed after the fact against a reference the resource never consulted. It is a property of the search itself, and the search engine's output can carry it: MaxQuant's `peptides.txt` includes a `Proteins` column listing every database entry that contains each peptide, so the compatible-source mapping the remedy in §5 asks for was **available to the search software**. We do not have IEAtlas's own `peptides.txt` or the code that transformed it into the public export, so we do not know when or how that mapping was lost; what the Methods establish is only that it was not propagated to the public atlas export, which records the ncORF attribution alone.

The exclusion test is also not a foreign operation. Building the library, IEAtlas reports that *"FASTA files of peptides were merged, and **peptides entirely contained within other peptides were removed**."* That is an exact substring-containment test — the same test this paper applies — run across the ncORF library to remove internal redundancy. It is not described as having been run **against the canonical proteome**, though that proteome was loaded into the same search. The remedy is that operation applied once more, in a direction the pipeline had already implemented.

The reading is falsifiable and we state the check: had the canonical half of that database been used to **exclude** shared sequences — as CrypticProteinDB and Raja et al. describe doing, and as their rates of 0.026% and 0.17% reflect — this rate would be near zero. It is 56.2% against our plausible-candidate reconstruction of the database (56.3% against the modern reference) — either way, not near zero, regardless of which exact release built the database.

This does not show that any peptide is canonically derived, and it does not show that anyone acted improperly. Sequence identity is symmetric, and the canonical entry is not "the right answer" either; MS identifies the sequence, never the locus. What it shows is that the evidence needed to mark these sequences *source-ambiguous* was inside the pipeline that produced the catalogue.

Under one pipeline, applied identically to every catalogue we could reprocess — same reference, same exact-substring criterion, same unique-sequence unit:

catalogue	exclusion rule in its Methods?	canonical-sequence overlap
CrypticProteinDB	**yes** — *"BLASTP... eliminate all proteins with alignment to canonical proteins"*	**1 / 3,810 = 0.026%**
Raja et al. (ovarian)	**yes** — *"peptides mapping to 'protein_coding'... were excluded"*	**5 / 2,979 = 0.17%**
IEAtlas	**not described** — *"only epitopes derived from non-coding regions were retained"*	**98,193 / 174,465 = 56.3%**

Table 2. Canonical-sequence overlap under one pipeline, one reference and one peptide unit, for two catalogues with an explicit exclusion rule versus IEAtlas, which has none described.

For context, Bedran et al. 2023 report residual canonical overlap of 1.4% (Erhard), 3% (Ouspenskaia), 4% (Chong) and 5% (Laumont). We do not compute a fold-change against those values. They were produced with a different reference, normalization, deduplication and peptide unit, and a ratio across pipelines is arithmetic rather than measurement. They are cited as published context. The controlled comparison is the table above.

It is not the ORF-class composition

IEAtlas's ncORF library explicitly includes pseudogenes, the class with the highest canonical compatibility. A peptide may carry more than one ORF label — 1,801 sequences (1.0%) map to more than one gene, and 546 carry both a pseudogene and a non-pseudogene ORF — so pseudogene and non-pseudogene sets are *not* complements. We instead report three mutually exclusive strata, which do partition the catalogue exactly:

stratum (mutually exclusive)	*n*	canonical-compatible
pseudogene-only	15,777	**59.4%**
non-pseudogene-only	158,142	**55.8%**
both labels (source-ambiguous within the atlas)	546	**93.0%**
total	**174,465**	**56.3%**

Table 3. The catalogue partitioned into three mutually exclusive ORF-class strata; canonical-compatibility is high in all three.

If the atlas contained no pseudogene ORFs whatever, the rate would be **55.8%**. Class composition moves the headline by half a percentage point. Separately, Raja et al. report 98 pseudogene-ORF peptides and **none** overlaps a canonical protein, because their exclusion rule removed those that did — same ORF class, different rule.

That 1.0% of sequences map to several genes *within the atlas's own annotations* is itself a small, direct instance of the ambiguity this paper is about.

R2. The consequence is observable inside the resource

An audit that stops at *"56.3% of these sequences are source-ambiguous"* invites the only question that matters: **so what?**

If a canonical-compatible sequence is in fact produced from the canonical locus it is compatible with, it would be expected to also be presented on **normal tissue**, because that protein is expressed there too. This is testable without resolving provenance: IEAtlas publishes its own normal-tissue epitope export (94,375 unique peptides), so the prediction can be checked **inside the resource**, comparing canonical-compatible sequences against the canonical-incompatible sequences of the same catalogue as a **within-resource comparator** (**Figure 2**).

IEAtlas cancer-catalogued sequences	also in IEAtlas's own normal-tissue export
canonical-compatible (98,193)	**22,003 = 22.4%**
canonical-incompatible — within-resource comparator (76,272)	6,976 = **9.1%**

Table 4. Canonical-compatible sequences are reported in IEAtlas's own normal-tissue export at more than twice the rate of canonical-incompatible ones.

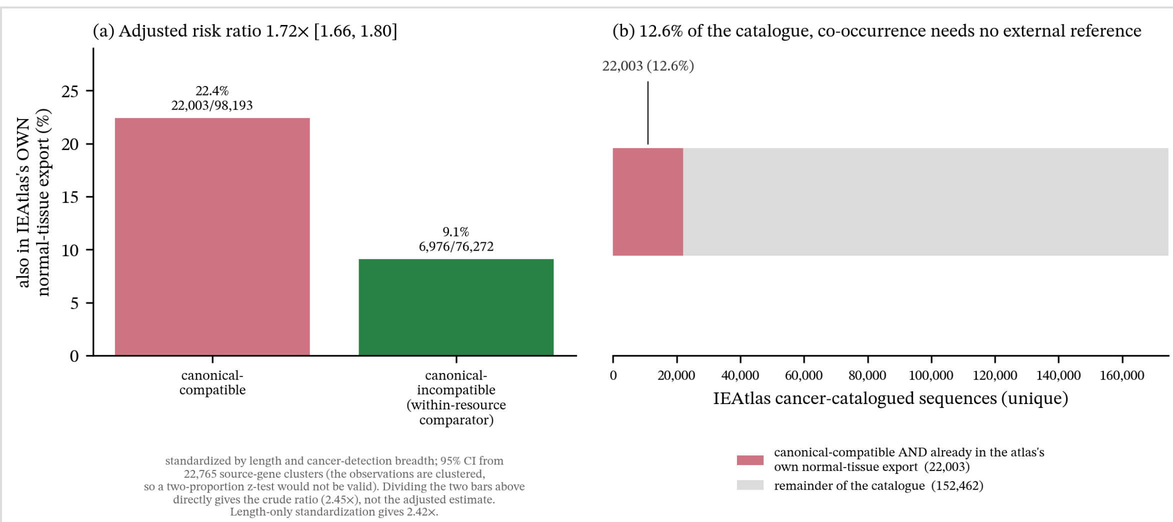

Figure 2. (a) The length- and cancer-detection-breadth-standardized risk ratio for co-occurrence in IEAtlas's own normal-tissue export, canonical-compatible versus canonical-incompatible cancer-catalogued sequences, with its gene-clustered bootstrap interval. (b) The 22,003 canonical-compatible sequences already in the normal-tissue export, as a share of the whole cancer catalogue — their co-occurrence in both of IEAtlas's own exports needs no external reference, though canonical-compatibility itself (R1) is assigned using Swiss-Prot.

Estimating the association, and naming what it can and cannot account for. These 174,465 sequences are *not* independent Bernoulli observations — they are plausibly clustered by source gene, gene family, dataset, tissue, donor, HLA allele and pipeline — and peptide length confounds both arms. Two of these axes are resolvable from IEAtlas's own published exports (source gene; tissue) and are modeled directly, below. Gene family, dataset, donor, HLA allele and pipeline are not joinable to individual peptide records in the files IEAtlas makes available (its sample-level metadata carries no key back to the peptide export) and are acknowledged, not modeled. Accordingly:

- **Per length, unpooled**, the risk ratio ranges **1.55–3.08** and the effect holds at **all 18** peptide lengths (8–25 aa).
- **Length-standardized** (direct standardization to the catalogue's own length distribution) the risk ratio is **2.42×** — essentially the crude 2.45×, so length is not driving it.
- **Length- and cancer-detection-breadth-standardized** (breadth bins 1, 2, 3, 4 and 5+ cancer types; common-support n = 174,413), the risk ratio attenuates to **1.72×**; gene-clustered bootstrap **95% CI [1.66, 1.80]**. Exact, unbinned breadth gives the same 1.72×. Thus measured generic detection opportunity explains part, but not all, of the association.
- **Gene-clustered bootstrap** (resampling 22,765 source-gene clusters with replacement, B = 2,000): **95% CI [2.32, 2.52]**.
- **Tissue-clustered bootstrap** (a second, independent clustering axis: resampling 15 cancer-type clusters with replacement, same B): **95% CI [1.86, 3.29]** — excludes 1.0 under this axis too, but read this interval as indicative, not precise: the 15 clusters are heavily unbalanced (the two largest hold 41% of the catalogue between them), giving an **effective cluster count of 7.9** (the Herfindahl measure $1/\Sigma p^2$) — below what cluster-bootstrap practice generally treats as reliable for percentile-CI coverage. It is reported as a second, cruder check, not as equal-weight evidence alongside the gene-clustered CI above.
- Within every ORF-class stratum, length-standardized: 2.39× (pseudogene-only), 2.40× (non-pseudogene-only), 1.75× (both, n = 546 — the smaller, source-ambiguous-within-the-atlas subgroup).

A further unresolved possibility is residual detection opportunity not captured by length and cancer breadth — including study/run opportunity, source-gene abundance, donor, allele and pipeline. The adjusted result is therefore an association after two measured standardizations, not a causal estimate of canonical origin or normal-tissue expression.

A two-proportion z-test is not valid for this contrast — it would treat a heavily structured catalogue as 174,465 independent experiments, and being directionally right would not rescue it. The clustered interval above is the correct measure of precision here.

A subset requires no inference at all — to see the co-occurrence, though not to name it "canonical-compatible." **22,003 unique sequences — 12.6% of the cancer catalogue — are both canonical-compatible and already present in the atlas's own normal-tissue export.** That these 22,003 sequences occur in *both* of IEAtlas's own exports is internal to the resource and needs no external reference; that they are *canonical-compatible* in the first place is R1's finding, and R1 uses Swiss-Prot. The two parts of the claim rest on different evidence — only the co-occurrence is reference-free.

A further caveat on what "present in the normal-tissue export" establishes: IEAtlas's Methods report that MaxQuant's *"'match between runs' option was set with default settings"* for this analysis, so an entry in the normal-tissue export is not guaranteed to be an independent MS/MS identification in that tissue — some fraction may be transferred (matched-between-runs) identifications from a run where the peptide *was* directly sequenced. IEAtlas's public exports do not carry a field distinguishing the two. Accordingly we say a sequence is **reported in**, or **co-listed in**, the normal-tissue export — not that it was independently *observed* or *presented* there — and the practical conclusion (normal-presentation review before treating it as tumour-restricted) is unchanged either way: a transferred identification still means the resource itself asserts the peptide's presence in that normal sample.

Staged external benign-tissue recurrence. We also cross-referenced the cancer catalogue against the 37 benign processed peptide tables deposited for PXD038782 (Hoenisch Gravel et al. 2023). **15,171** IEAtlas cancer sequences recur in those tables; **15,170** are exact canonical-compatible. Of these recurrences, **6,244** are absent from IEAtlas's own normal export, while 7,035 recur across at least two deposited tissue labels. This is external sequence-list corroboration outside the two IEAtlas exports, but it is deliberately not called direct validation: PXD038782 is a partial submission, its deposited tables come from a canonical-proteome PEAKS DB search rather than joint canonical/ncORF competition, and the available metadata do not support donor-allele matching or USI-level verification. The result establishes benign processed-table recurrence of the sequence, not its source locus or target-safety consequence.

What this means, bounded. This is **consistent with, but not specific to**, greater detectability or expression of canonical-compatible sequences. It does not show that any individual sequence is canonically derived: inclusion in a normal-tissue export is evidence about **reported detectability**, not about **source** — and, per the MBR caveat above, not necessarily about direct observation in that tissue either. Its practical consequence is that such a sequence warrants normal-presentation review before it is treated as a tumour-restricted target — clinical risk depends additionally on allele matching, tissue context, abundance and TCR avidity, none of which we assess. Concretely: IEAtlas's own sample metadata records a Class I/Class II split among its contributing mass-spectrometry runs (1,942 Class I, 1,159 Class II), but that field is not joinable to individual catalogued peptides in the **bulk exports used in this study** — so neither the 56.3% overlap nor the 22,003 figure can be stratified by HLA class or allele from the files we analyzed, and we do not attempt it. (IEAtlas's live gene-search interface does expose HLA class and allele as per-epitope, per-query fields; that is a different access path from the bulk exports this paper's pipeline consumes, and re-deriving 174,465 class/allele labels one query at a time is out of scope here.)

R3. Latent canonical ambiguity differs enormously between libraries — and does not compose

Applying the same measurement to the **search space** rather than the output — the distinct 9-mers of each ncORF library, and how many also occur in reviewed canonical human proteins (**full libraries, no sampling**; **Figure 3**):

ncORF library	ORFs	distinct 9-mers	canonical-compatible
nuORFdb v1.2 — integrated by IEAtlas	229,251	8,448,245	**34.1%**
GENCODE Ribo-seq ORFs (phase 1)	7,264	245,094	**2.4%**
GENCODE Ribo-seq ORFs (phase 2)	28,359	502,528	**1.0%**

Table 5. Latent canonical ambiguity (9-mer canonical-compatibility) of each ncORF library, measured under one pipeline.

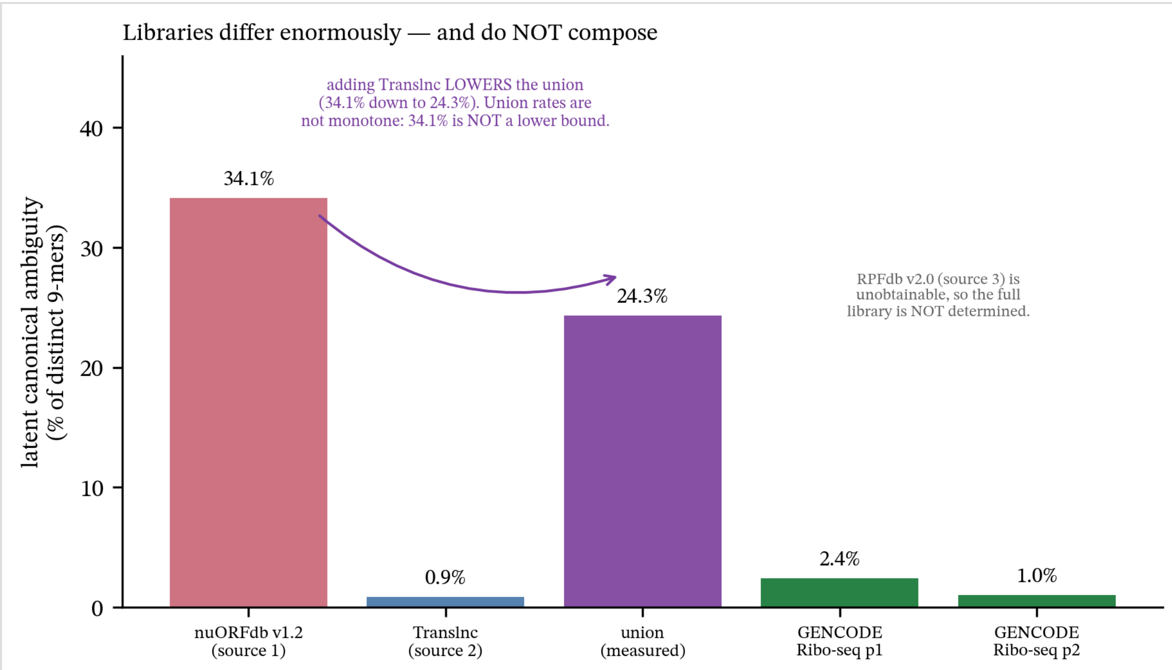

Figure 3. Latent canonical ambiguity differs enormously between ncORF libraries and does not compose — nuORFdb union Translnc (human-only) falls below nuORFdb alone.

Both libraries were measured by us, under one pipeline, so the contrast *is* a controlled one: **ncORF libraries differ by 14–34× in latent canonical ambiguity.** Whole-ORF containment is low throughout (0.2–0.8%), so this is **extensive partial sharing**, not whole ncORFs nested inside canonical proteins.

Independent corroboration. These figures were obtained twice, by separate implementations over different k-mer windows: an 8–11mer candidate-universe enumeration gives nuORFdb **34.4%** and GENCODE Ribo-seq (phase 1) **2.5%**; the 9-mer enumeration above gives **34.1%** and **2.4%**.

The integrated library does not inherit nuORFdb's ambiguity

34.1% is **not a lower bound** on IEAtlas's complete integrated library (nuORFdb + RPFdb + Translnc). Adding a library B to nuORFdb A changes the rate to $|(A \cup B) \cap C| \;/\; |A \cup B|$, which is **not monotone**: if B contributes mostly non-canonical k-mers, the combined proportion **falls**.

It falls. We obtained Translnc — a second of IEAtlas's three sources, in the version it cites — and measured the union directly. **Translnc's distributed FASTA is multi-species**: 435,173 human (ALL-CAPS gene-symbol convention) headers followed by 148,667 mouse-convention headers (`Tug1-...`, `Gm10619-...`, RIKEN clone IDs). A mouse 9-mer cannot legitimately enlarge the denominator of a rate scored against a *human* canonical reference — it dilutes the calculation with sequences a human immunopeptidome search space never contained. We report the **human-only** union as primary and the whole-file union alongside it as a diagnostic of how much species-filtering matters:

library	distinct 9-mers	canonical-compatible	rate
nuORFdb v1.2	8,448,245	2,884,119	**34.1%**
Translnc, human-only entries	3,728,231	32,096	**0.9%**
nuORFdb ∪ Translnc (human-only)	11,934,337	2,902,793	**24.3%**
(diagnostic) Translnc, whole file (human + mouse)	*6,164,584*	*39,382*	*0.6%*
(diagnostic) nuORFdb ∪ Translnc, whole file	*14,364,357*	*2,907,391*	*20.2%*

Table 6. The nuORFdb/Translnc union: adding a second library lowers, not raises, the combined ambiguity rate.

Translnc is almost free of latent canonical ambiguity under either filtering, and the two libraries are nearly disjoint (2.0% of the human-only union's 9-mers occur in both). Adding it therefore enlarges the denominator far faster than the numerator, and the rate **drops by 9.8 pp, from 34.1% to 24.3%** (human-only) — **13.9 pp, to 20.2%**, if mouse sequences are left in undiluted. This is the non-monotonicity above made concrete: adding Translnc **lowers** the rate, on one of IEAtlas's own other sources, under either convention. (Re-scored against Swiss-Prot 2022_01, human-only union: 24.3%, essentially unchanged.)

The gap between the two is plausibly not arbitrary. Translnc catalogues ORFs annotated on **lncRNA transcripts**; nuORFdb catalogues ORFs of **coding genes** — uORFs, dORFs, out-of-frame and in-frame alternative ORFs — whose reading frames overlap or abut the very proteins in the canonical reference. lncRNA annotation is not a guarantee of zero coding-gene overlap — lncRNA loci can be intergenic, antisense, intronic or sense-overlapping with a coding gene, and we have not stratified Translnc by that relationship — but the measured **~40×** gap between the two libraries (34.1% vs 0.9%, human-only) is consistent with a library's latent ambiguity being **contributed to** by whether its ORFs sit inside coding genes, a property its builders already know and could report at zero cost, without our claiming that relationship as a general mechanism verified by direct genomic-context stratification. This is the concrete content of recommendation (c) in §5.

What is and is not now bounded. RPFdb v2.0 remains genuinely unavailable — the live site serves only v3.0, and it distributes RibORF genomic *coordinates*, not amino-acid sequences, so back-translation would produce *our* library rather than IEAtlas's. Carrying its contribution as the single unknown m (novel 9-mers it adds, of which x are canonical), the combined rate is $(h + x) / (u + m)$. Maximising over both, the three-source library's 9-mer ambiguity **cannot exceed 53.5%** under the human-only union (**47.6%** under the whole-file union), distribution-free and whatever RPFdb contains. That cap is a **hard ceiling, not an estimate**: it is attained only in the corner where every canonical 9-mer not already in the union is contributed by RPFdb *and* RPFdb contributes nothing else. There is **no positive lower bound derivable from the sources we hold**: as m grows, $h/(u + m) \to 0$. The combined rate is not determined by our measurement, and no measurement of a subset of the libraries could determine it.

The library is not sufficient on its own. The evidence that the *exclusion step*, not the library, is what governs a catalogue's overlap rate is our own same-pipeline reprocessing: CrypticProteinDB and Raja et al., which describe explicit exclusion rules, sit at **0.026%** and **0.17%** (§1). Ouspenskaia et al. add the one control neither of those provides — they searched the **same nuORFdb** and their published catalogue is nonetheless low (3%, Bedran et al.). We attach **no arithmetic** to that 3%: it is a published cross-pipeline figure, and the unit caveat above applies to it as much as to any other. The qualitative point is all we need and all we claim — a high-ambiguity library does not force a high-ambiguity catalogue.

A detection effect, tested — scope of inference

We do not compare the catalogue's rate to the library's as levels. The catalogued 56.3% and nuORFdb's 34.1% are different objects: 56.3% is over distinct catalogued *peptides* at native lengths, after search, FDR and deduplication; 34.1% is over distinct *9-mers* of a candidate search space in which nothing has been detected. Different units, different denominators, different lengths — the same cross-unit error as the fold-change comparison against other catalogues that §1 declines to make, and it is not rescued by the union figure either (56.3% must not be read against 24.3%/20.2%, or against the 53.5%/47.6% caps).

The detection hypothesis is nonetheless real and testable *without* that comparison — a peptide of an abundant, ubiquitous protein should be over-detected in an immunopeptidome relative to its share of the search space. Both tests below are internal to a single space or are ratios of ratios, so neither requires a cross-unit subtraction.

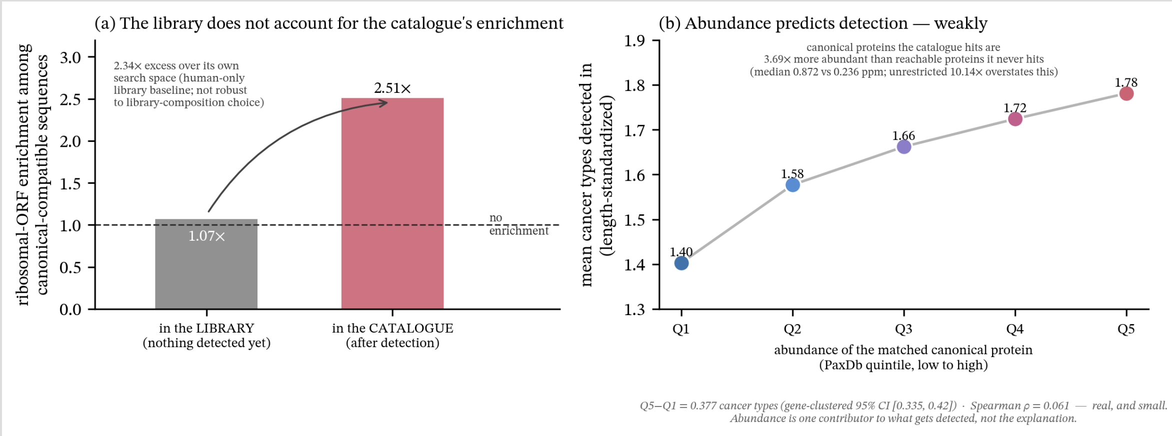

Figure 4. (a) Ribosomal-ORF enrichment among canonical-compatible sequences, in the library that was searched versus in the resulting catalogue — an excess over the human-only library baseline that is not robust to library-composition choice (see text). (b) Abundance predicts detection breadth, weakly — the Q1-to-Q5 quintile trend after length standardization, with its gene-clustered bootstrap interval.

Prediction 1 — breadth of detection. A peptide of an abundant, ubiquitous protein should be detected across more of IEAtlas's 15 cancer types. Canonical-compatible sequences are seen in a mean of **1.62** cancer types versus **1.33** for canonical-incompatible ones, and **28.3%** appear in ≥2 types versus **16.8%**. Because short peptides both match canonical proteins more readily and recur more often, we stratified by length: the effect holds in **18 of 18** length strata (8–25 aa). It is not a length artifact.

Prediction 2 — enrichment for the abundant housekeeping class. Ribosomal proteins are the textbook abundant, ubiquitous class. In the catalogue, canonical-compatible sequences are **2.51**× enriched for ribosomal-gene ORFs (2.89% vs 1.15%). This could be pure library composition, so we measured the same enrichment **in the library**, over **distinct 9-mers** — the same unit as the library's headline ambiguity rate (§R3), not raw (ORF, 9-mer) occurrence pairs, which would double-count a 9-mer once per ORF that contains it and so are not the same unit as the catalogue side. There, ribosomal-associated 9-mers are **depleted** among canonical-compatible candidates in the nuORFdb-only library (**0.69**×), giving a nuORFdb-only-baseline excess of **3.66**× — but nuORFdb alone is not the full three-source library IEAtlas integrates, and this specific figure is not robust to that choice (below). Folding in Translnc — a second source we hold, whose union with nuORFdb we already computed for the headline ambiguity rate (34.1% → 24.3%, human-only) — moves the library-side ribosomal rate up and the excess down to **2.34**× under the human-only convention we report as **primary throughout this paper**, or **1.85**× using the whole Translnc file with mouse sequences left in (the diagnostic convention, kept only to show what species-filtering changes; **Figure 4** plots the human-only, primary pair). RPFdb v2.0's contribution is, as in §R3's headline ambiguity rate, unmeasured. The qualitative conclusion — the catalogue's enrichment exceeds its own search space's, under every variant we can compute — is what P2 supports; **the specific 3.66× should not be quoted as the headline number, and neither figure should be read as showing the excess "arose during detection" in some further causal sense beyond that comparison.**

Note what Prediction 2 does and does not compare. It is a **ratio of ratios** — an *enrichment* measured within the catalogue, set against the *same enrichment* measured within the library, both over distinct sequence units — and a ratio of ratios is dimensionless, so it survives the unit mismatch that would make a direct comparison of the catalogue's and the library's raw ribosomal-association rates meaningless. But dimensionless is necessary, not sufficient: the two ratios must also be computed over the **same observational unit**, or the excess can reflect how each side weights individual sequences rather than any real effect — a 9-mer repeated across many near-identical ORFs (ribosomal paralogs are exactly this) would otherwise be counted once per occurrence rather than once per sequence, inflating its apparent weight. The catalogue side is already over distinct catalogued peptide sequences (as throughout this paper); the library side must therefore use **distinct 9-mers**, not raw (ORF, 9-mer) occurrence pairs, to match — the same unit as §R3's headline library rates. It says the association between being canonical-compatible and being ribosomal is stronger in the catalogue than in the search space it was drawn from. It does **not** license any claim that the catalogue's overlap rate sits some number of percentage points above the library's — that quantity is not defined.

A further check on what "ribosomal" is doing here. `RPL*/RPS*/MRPL*/MRPS*` gene symbols include not only protein-coding ribosomal genes but their retro-transposed pseudogenes (e.g. *RPS3AP12*, §S1) — canonical-compatible for an unrelated, already-established reason (pseudogene biology, not abundance or detection). Splitting the class with the same authoritative NCBI pseudogene→parent registry used for S1 shows the excess is not pseudogene contamination wearing a ribosomal label: isolating true protein-coding ribosomal genes alone gives an excess of **3.47**× (nuORFdb-only baseline) — at least as large as the **3.66**× pooled figure above — and the ribosomal-named-pseudogene subset separately shows a smaller but still positive excess (**2.16**×), consistent with those sequences riding on their parent's abundance rather than a distinct mechanism.

Prediction 3 — the direct test. Predictions 1 and 2 use *breadth of detection* and *ribosomal membership* as **proxies** for abundance. We replace them with a measurement: **PaxDb v6.1** (human, whole-organism integrated, ppm), joined to the canonical proteins the catalogued sequences actually match (**91.4%** of the reviewed proteome joins; 96,210 of the 98,193 overlapping sequences carry an abundance).

Which canonical proteins do the catalogued sequences hit? **Abundant ones.** At the protein level — where there is no peptide-clustering problem at all, because the unit *is* the protein — canonical proteins hit by a catalogued sequence have a median abundance of **0.872 ppm** against **0.086 ppm** for those never hit, a **10.14**× difference (AUC **0.679**).

The "never hit" population is not restricted to proteins the search could plausibly have reached. Splitting it by whether the protein's gene appears anywhere else in the 174,465-sequence catalogue — a candidate existed for the gene, it just didn't overlap this specific protein — against genes absent from the catalogue entirely: restricting the comparison to the **reachable** subset (n = 3,480, median 0.236 ppm) drops the fold to **3.69**× (AUC **0.601**); the fully unreachable subset (n = 3,562, median 0.042 ppm) alone gives an even larger 20.76× (AUC 0.755). The direction is unchanged under every split — hit proteins are more abundant than any comparator — but the unrestricted 10.14× overstates it by roughly this much; **3.69×** is the more defensible figure. This selection effect does not touch the detection-breadth test below, which never uses the "never hit" population.

Reachability by catalogue co-occurrence is one operationalization; reachability by library content is another, and gives a different, weaker restriction. A protein's gene appearing elsewhere in the catalogue is an *output*-side signal. A more search-side definition asks whether the reconstructed ncORF library itself (nuORFdb ∪ Translnc, human-only, **11,934,337** distinct 9-mers) shares even a single 9-mer with the protein at all — i.e., whether the search space contained *any* sequence that could in principle have been proposed as an overlapping candidate, independent of what the catalogue happened to surface. Under this definition, **3,912** of the 7,042 never-hit proteins qualify as library-reachable (median **0.100** ppm) against **3,130** that share no 9-mer with the library at all (median 0.070 ppm) — dropping the fold to **8.72×** (AUC **0.667**), a smaller correction than the catalogue-co-occurrence restriction. This is expected, not a discrepancy: with **12.1 million** distinct 9-mers in the reconstructed library, most proteins of ordinary length share *some* 9-mer with it by chance, so "shares ≥1 library 9-mer" is a comparatively weak filter — only 44.4% of never-hit proteins are excluded by it, against 50.6% excluded by the catalogue-co-occurrence restriction. That the stricter, output-based restriction (3.69×) sits *below* the library-content one (8.72×), rather than above it, means the originally reported figure was already the more conservative of the two available corrections, not an outlier chosen for effect. We report both rather than picking one.

Does abundance predict how broadly a peptide was detected? Yes — but weakly, and we state the weakness rather than the headline. Binning canonical-compatible sequences by the abundance of their matched canonical protein, mean detection breadth rises across every quintile after length standardization, from **1.40** cancer types (Q1) to **1.78** (Q5) — a Q5–Q1 gap of **0.377**, with a gene-clustered bootstrap **95% CI [0.335, 0.42]** that excludes zero under **both** cluster definitions (matched canonical gene, and IEAtlas source gene), and which reproduces on the previous PaxDb release (**0.38**). The crude, unstandardized trend saturates — it climbs through the low quintiles and flattens across the top — and is monotone *only* after length standardization; the crude gap is **0.23**. The direction and the interval are solid. **A strong dose–response is not**, and we do not claim one.

Three controls, each of which could have killed it:

control	the attack it answers	result
peptide length	short peptides both match canonical proteins more readily *and* recur more	trend holds in **18 / 18** length strata
protein length	longer proteins are hit more often *by chance*, so "hit" may just mean "long"	hit proteins **are** longer (median 490 vs 360 aa) — but the abundance effect holds in **10 / 10** protein-length deciles
placebo	does the machinery invent trends?	breaking the peptide→protein link **collapses** the gap to **0.0** (0 of 200 draws reach the observed 0.377)

Table 7. Three controls for the abundance-predicts-detection trend, each capable of falsifying it.

What this licenses, and what it does not. The abundance explanation is now **measured, not proxied**: canonical-compatible sequences preferentially match abundant canonical proteins, and that abundance predicts detection breadth, surviving peptide length, protein length and a placebo. But the per-sequence association is **weak** (Spearman ρ = **0.061**): abundance is *one* contributor to which sequences get detected, **not** the whole explanation, and we do not claim otherwise. Nor does any of this speak to provenance — it is evidence about **what gets detected**, and MS still identifies the sequence, never the locus.

R4. The additive statistical problem, and the remedy the field already demonstrated

The phenomenon is not novel (Woo et al. 2014 measured it empirically; Choi & Paek 2024). Nor is the bound's general form: a worst-case interval on one unidentified component of a convex mixture is the same structure as Manski's classical partial-identification bounds (Manski 1989). We checked directly and confirmed that the closed-form instantiation for class-conditional FDR below is not already available, verified against Woo et al. 2014, which does not contain it. Stated here because it is *additive* to R1–R3, and because IEAtlas reports nothing that would allow it to be assessed.

From a reported pooled FDR q and class fraction f, the class-specific FDR is only **set-identified**. (The Supplement derives this exactly for the *realized* false-discovery *proportion* — writing it as *FDP*, not *FDR*, to keep the arithmetic from reading as a claim about an expectation — and evaluates it at the reported q as a stand-in for the true, unobserved proportion. We use q and *FDR* here as the field's working shorthand for that same estimated quantity.)

$$\Theta\_N(q, f) = [\ \max(0, (q - (1 - f)) / f), \min(1, q / f)\ ]$$

From q and f alone, this identified set is sharp. IEAtlas reports q = 0.05 and 245,870 non-canonical epitopes, but **no canonical count**. f is therefore unknown and the interval is **unconstrained**.

The 245,870 figure is also not the right unit for *f*, independent of the missing canonical count — and it is not even the quantity we first took it for. IEAtlas's Methods report q = 0.05 as a **peptide-spectrum-match (PSM)** false-discovery rate — *"A peptide spectrum match false discovery rate (FDR) of 0.05 was used, and no protein FDR was set"* — a quantity estimated over accepted PSMs. 245,870 is a deduplicated epitope-level total, but it is **not** a cancer-plus-normal-tissue union at the peptide level. IEAtlas's Methods separately report that *"54 017 epitopes for HLA-I and 51 015 epitopes for HLA-II passing HLA immunogenic tests... account for 37.16% and 50.76% of HLA-I- and HLA-II-binding epitopes, respectively"* — an unambiguous pairing (the denominator is named explicitly) from which the HLA-I and HLA-II totals are 54,017 / 0.3716 ≈ **145,363** and 51,015 / 0.5076 ≈ **100,502**. These sum to **245,865**, matching the published 245,870 to within the rounding of two published percentages. IEAtlas's own public browse interface independently states the exact totals directly, without the rounding: **145,366**

HLA-I-bound and 100,504 HLA-II-bound non-canonical epitopes, which sum to **exactly 245,870** — confirming the Methods-derived estimate above by a second, independent route. (IEAtlas's Methods elsewhere also state that *"60.60% and 41.90% non-canonical epitopes were bound by HLA-I and HLA-II allotypes, respectively"*; we do not use this sentence in the arithmetic above, because its denominator is not stated explicitly and multiplying it by 245,870 gives totals — 148,997 and 103,020 — that do not agree with the two totals above closely enough to be the same quantity. It is left unreconciled and not load-bearing for what follows.) **245,870 is, at minimum, extremely close to IEAtlas's HLA-I-bound-epitope count plus its HLA-II-bound-epitope count — a class-summed total, not a tissue-summed one.** It might appear that 174,465 + 94,375 − 245,870 = 22,970 gives the number of sequences shared between the cancer and normal exports; that arithmetic is correct but the inference is wrong. The two exports' true peptide-level overlap is not an algebraic residual of a class-summed total — it is directly measured, elsewhere in this paper (R2): **22,003 + 6,976 = 28,979** cancer-catalogued sequences also occur in the normal-tissue export, roughly 6,000 more than the residual implied. Because 245,870 conflates HLA class with nothing about tissue, it cannot be used to check, or substitute for, the cancer/normal totals at all; this is a second, independent unit mismatch, not a restatement of the PSM-versus-peptide one above. Neither the number of accepted PSMs nor the non-canonical share of accepted PSMs — the quantities that would instantiate f at the same unit as q — is published anywhere we have found. Closing Θ_N(q, f) would require PSM-level, class-resolved accept/decoy counts, not merely a peptide-level canonical count, at whatever tissue or class aggregation.

Reporting the per-class accepted target and decoy counts (*T_N*, *D_N*), the threshold, the unit and the convention would make the selected class-specific target–decoy estimate **reconstructible**. This is not the only information that could tighten the interval — calibrated class-specific posterior error probabilities or entrapment measurements could also do so — and *D_N* does not identify the true class-specific false-discovery proportion. It is simply the cheapest sufficient object, and one the pipeline already computes.

Reconstructibility is not statistical control. The ledger is a necessary audit object, but naïvely running target–decoy competition independently inside each class does not by itself guarantee control of the combined error criterion. Group-walk was developed for precisely this setting and supplies a rigorous group-aware procedure (Freestone et al. 2022). We therefore recommend the ledger for auditability and a grouped procedure, or an empirically validated class-specific protocol, for control.

The field has demonstrated the remedy, and its cost. Ouspenskaia et al. searched a combined annotated-ORF/nuORF database and reported that a 1% global FDR gave *"4.6% overall, and as high as 14% for 3′ dORFs"* among nuORF peptides; group-based filtering *"removed 24% of nuORF peptides overall, and up to 76% of peptides assigned to 3′ overlap dORFs."* **The 4.6% is a pre-correction diagnostic, not the error rate of their published catalogue.** They are a positive exemplar.

R5. Two distinct problems; one reporting remedy; and what it costs

The two defects are **complementary, not the same**, and merging them is a category error:

- **Source ambiguity (R1–R3).** The peptide is *correctly identified as a sequence*; its **source locus** is unresolved. This is not an FDR problem — no identification is wrong.
- **Class-specific FDR under-control (R4).** A pooled threshold *can* under-control the minority class, meaning some ncORF identifications *may be* wrong — the spectrum was not that peptide. For IEAtlas specifically, R4 shows this cannot be checked either way: with *f* unknown, $\Theta\_N(0.05, f)$ reaches down to 0, so the published numbers are equally consistent with zero wrong ncORF identifications as with many. The mechanism is established; whether it bites for this resource is not.

It might seem that FDR could not *explain* the 56.3%, on the grounds that a composition-matched shuffle places chance canonical overlap near 0.1% — but that argument does not hold. A false target PSM is not an arbitrary shuffled string; it is an accepted, incorrect candidate drawn from the actual search database, and its class composition is not described by a shuffle null. The correct statement needs no such argument: source ambiguity is present *even when the sequence is correctly identified*. FDR concerns whether the spectrum was assigned to the right sequence; canonical overlap concerns whether that correctly-identified sequence determines a source. They are different objects.

Aggregation changes the recurrence estimand

The same projection problem appears on a second axis: **a recurrent peptide sequence is not necessarily a recurrent peptide–HLA target**. We tested this in a separate, frozen audit of the 2025-07-07 ImmunoVerse supplementary tables (Li et al. 2025), chosen because its current repository also publishes scan-level outputs that make a bounded reconstruction possible. Table S7's 451 used biological labels all joined exactly to Table S3, which contains 47 published source-study identifiers. Among the 1,000 most sequence-recurrent peptides, the median recurrence fell from **11 sample/condition labels to 7 source studies**. Even after conservatively bounding the six labels that map to more than one study, the median ratio of sequence-study recurrence to recurrence of the best predicted peptide–HLA was **1.25×**, and **72.2%** of ratios exceeded one (Supplement S3).

A current raw-linked DLBC subset then supplied a direct worked example. Two sequences, `AEGPDHHSL` and `VPHTRPVSL`, had at least one current `Identified = +` scan in every cell line listed for that sequence in the historical catalogue. The reported HLA genotypes of those cell lines had an empty intersection. Conditional on the published genotypes, one fixed peptide–HLA allomorph therefore cannot explain the recurrence across all listed contexts: **the sequence recurs; one molecular target does not**. This is a two-peptide demonstration, not a prevalence estimate, and it does not assign the presenting allele within any sample.

ImmunoVerse is a positive provenance-recovery example, not an allegation of irretrievable evidence loss: its current release publishes MaxQuant, rescoring and consolidated scan outputs separately from the peptide catalogue. In a BLCA positive-control subset, all **16 / 16** historical sequences and all **18 / 18** scan rows implied by the catalogue's PSM counts were recovered exactly. The point is that the catalogue row alone does not carry the joins; the separately published evidence chain allows them to be rebuilt.

An atlas-level provenance-retention profile addresses both.

(a) Per peptide — an exclusivity flag and all compatible source loci. Consistent with MIAIPE and mzIdentML, state whether the sequence is unique to the nominated ncORF within the searched space, and if not, list every compatible source. This preserves the peptide *and* the ambiguity, rather than discarding either, and it is strictly more informative than the exclusion rule it generalises.

This is primarily a retention problem, not a request for a new exotic data structure: search outputs can already preserve the mappings. A remedy that demanded an impractical long list of loci per peptide would nevertheless be glib, so we measured the list. Across the 97,999 catalogued sequences that match the canonical half of IEAtlas's own search database, the **median number of compatible canonical genes is 1**; **93.1% are compatible with exactly one** canonical gene and **98.0% with at most two** (maximum 22). The label is, for the large majority of ambiguous sequences, **a single gene symbol**. And as §1 showed, a concatenated search already computes it: MaxQuant's `peptides.txt` lists every database entry containing each peptide. The remedy asks resources to preserve mappings their own pipelines already produce.

(b) Per class — a class-decoy ledger. Accepted target and decoy counts per class, the class definitions, the thresholding stage, and the formula used — enough to reconstruct the class-specific estimate. Ouspenskaia et al. demonstrate it is achievable.

(c) Per library — publish the latent canonical ambiguity. One number (R3), cheap to compute, and it tells every downstream user how much an exclusion rule matters for the library they are about to search. No ncORF library currently publishes it.

The cost, stated plainly. Under a definition that requires canonical-incompatibility, **98,193 of 174,465 unique cancer-catalogued sequences (56.3%)** would be ineligible for designation as uniquely non-canonical. That does **not** require deleting them from the atlas — the remedy in (a) is to retain them and label them source-ambiguous with their compatible loci. For scale, class-specific FDR control cost Ouspenskaia et al. 24% of their nuORF peptides and up to 76% of one ORF class. Applying an established criterion to an ncORF catalogue is expected to be expensive. That is not an argument against applying it — it is an argument for applying it before, rather than after, a candidate enters a clinical pipeline.

Discussion

Every resource examined describes its own procedures accurately and in public; that is the only reason this audit was possible at all. Two of the catalogues measured here already apply an exclusion rule. Ouspenskaia et al. already solved the statistical half and published the cost of doing so. The field knows how to do this. What had not happened, as far as we have been able to determine, is a check of whether the resources already in use apply it. The observed overlap is compatible with several mechanisms — database composition, annotation transfer, search-space design — and sequence identity is symmetric: MS identifies the sequence, never the locus, so this analysis does not show any specific ncORF antigen to be non-real. Distinguishing among these mechanisms for any individual entry requires the original search outputs and peptide-level provenance.

The unifying issue is therefore not merely label accuracy but **whether a confidence statement survives a change of unit**. PSM filtering controls a spectrum-assignment estimand under its stated procedure. Alternative-mapping removal, sequence deduplication, source-label projection and recurrence counting then create source, catalogue and target-recurrence claims. Those downstream claims do not inherit PSM-level calibration automatically. Our data establish two concrete failures of automatic inheritance—source attribution in IEAtlas and one-pHLA recurrence in the bounded ImmunoVerse example— while R4 establishes that IEAtlas's public output does not permit the class-specific error behaviour to be reconstructed. We do **not** show that atlas aggregation empirically increased IEAtlas's FDR; that question requires class-resolved target/decoy data or a valid entrapment experiment.

A prospective falsification test follows directly from this distinction. TIPs, published during this revision, learns a sample-specific transposable-element target database from de novo sequence tags and reports stringent FDR control (Wu et al. 2026). In the pinned public implementation, TE target entries are selected using spectra from the sample before reversed search decoys are appended. That ordering creates a testable calibration question: do targets and decoys experience equivalent effective selection opportunity through the complete pipeline? It is **not** evidence that TIPs's reported FDR is inflated or that any reported candidate is false. We therefore preregistered a within-condition, held-out benchmark using metadata-validated 0D5P biological units, together with entrapments introduced before database selection. Same-data versus independently learned database yield is a transfer endpoint, not an FDP estimate; any error-calibration claim requires an entrapment estimator whose assumptions pass a prespecified simulation gate (Wen et al. 2025). Until that benchmark runs, TIPs is a motivation and falsification target, not evidence of a confidence-propagation failure.

Where the ambiguity comes from, and where the remedy belongs. A third of nuORFdb's peptide space is canonical by sequence. We cannot say what the corresponding figure is for IEAtlas's full integrated library, and we do not claim the library *quantitatively explains* 56.3%. But a library that publishes its own latent ambiguity lets every downstream group know in advance how much an exclusion rule will cost them, and the detection effect measured in R3 shows that what a catalogue reports is not simply a readout of what its library contains. We have not found this number published for any ncORF library, and computing it is inexpensive — a distinct-9-mer overlap scan against a reference proteome, as done here.

The stakes are not abstract. A sequence catalogued as a cancer epitope, which is canonical-compatible and which the same atlas's own normal-tissue export already lists, is not a promising tumour-restricted target without further review. **22,003 unique sequences in IEAtlas meet both conditions.** (R3 shows canonical-compatible sequences skew toward abundant canonical proteins on average, but that association is population-level and weak per sequence, $\rho = 0.061$ — we do not additionally filter this figure by abundance, and do not claim each of the 22,003 individually matches an abundant protein.) We are not claiming that any of them *is* canonically derived — MS cannot say — but a target-selection pipeline that cannot distinguish them is selecting under an ambiguity it has not been told about.

We looked for a confirmed instance of this and, on direct verification, did not find one. Choi & Zhang (2025)'s PepQueryMHC, for example, maps candidate peptides directly against translated RNA-seq reads and independently filters them against the Human Protein Atlas; it cites IEAtlas's peptide counts only for scale, and cross-checks that its own top candidates are absent from IEAtlas's normal-tissue export as one more piece of corroborating evidence — not as the source of its canonical/non-canonical calls. BamQuery (Ruiz Cuevas et al. 2023) makes a related choice more generally: independent of which catalogue a candidate peptide comes from, it assigns origin from RNA-seq expression directly rather than inheriting a source label. We are not aware of a confirmed instance of a downstream group reusing an ncORF catalogue's source labels without an independent check of its own, which is reassuring but does not make the underlying risk hypothetical: a catalogue that does not flag source ambiguity still requires every downstream user to re-derive the check PepQueryMHC happened to run, rather than inherit it.

Limits, stated plainly.

- The overlap is **reference-relative** — it is $N(R)$. For a fixed query set, overlap is monotone under **nested expansion of the same *R***; 56.3% is a lower bound with respect to supersets of *this* reference, **not** with respect to every possible reference definition. A narrower or differently defined reference can lower it. The era-correct check (56.2% against Swiss-Prot 2022_01) is the one that matters for judging the resource at build time.
- **No individual peptide's provenance is resolved.** Sequence identity is symmetric.
- We hold nuORFdb and Translnc but **not RPFdb v2.0**, which is unobtainable. The latent ambiguity of IEAtlas's complete integrated library is **not determined** — capped at 53.5% (human-only Translnc union; 47.6% under the whole-file diagnostic), with no positive lower bound derivable from the sources we hold, and certainly **not bounded below by 34.1%**.
- The 1.4–5% values for four other catalogues are **published figures from a different pipeline**, cited as context. We report **no fold-change** against them.
- Class labels for non-pseudogene classes are **source-supplied and uncorroborated**.
- The pseudogene→parent homology analysis (Supplement) uses a **curated, versioned** annotation (NCBI Gene `gene_group`) rather than symbol-stripping,

and the parent hit survives a **family-respecting** null even when the decoys are the parent's own **strong paralogs** (52.3% observed vs 16.6%, $p < 1e\text{-}4$) — addressing, respectively, the concern that a symbol heuristic could be wrong and the concern that a naive permutation ignores gene-family structure. It stays in the Supplement anyway: **133 testable peptides**, and the curated mapping is not independent of the symbol it replaces, because HGNC names a pseudogene *after* its parent. It explains *why* part of the pseudogene class is source-ambiguous; it does not measure the headline.

- The detection-bias result (R3) is now a **direct measurement** (PaxDb v6.1), not a proxy, and it survives peptide length, protein length and a placebo. But the **per-sequence association is weak** (Spearman $\rho = 0.061$). Abundance is **one** contributor to what gets detected, not the explanation.

Evidence required for direct verification. If IEAtlas's pipeline resolves source attribution in a way its Methods do not describe — for instance through a protein-inference step that assigns shared sequences to the canonical protein — then the 56.3% has an innocent explanation, and this paper is a correction to a misreading of the Methods rather than a finding about the resource. That step would have to be deliberate and purpose-built, not a default outcome of the search: a 2026 TransCODE consortium paper implements exactly this kind of explicit exclusion for HLA immunopeptidomics with its own numeric criteria, which it describes as *"less strict than those used by PeptideAtlas to avoid mapping canonical protein derived peptides to ncORFs"* (Deutsch et al. 2026) — naming a stricter scheme built for exactly this purpose, language that only makes sense if the underlying search process does not do this on its own. IEAtlas's Methods describe no equivalent step, and its sole supplementary file (four figure captions and their associated images; no methods text) describes none either. Publication of these search outputs (`peptides.txt`: `Proteins`, `Leading razor protein`, `Unique (Proteins)`) would permit direct verification of this specific alternative.

Methods

Reference

Canonical human proteome: UniProt/Swiss-Prot reviewed human sequences (R) — **20,431 entries, 11,418,104 residues**, fetched live via UniProt's REST API (`reviewed:true AND organism_id:9606`; the query is unversioned, so the SHA-256 hash printed by `reference_provenance.py` and recorded in `data/SOURCES.md` is the reproducibility anchor, not a dated release string). All overlap statistics are `N(R)` and are reference-relative. IEAtlas's Methods state only that the canonical proteome was "downloaded in February 2022" — a month, not a version string, and we do not treat it as identifying a single release. UniProt's release cadence in that window was roughly quarterly: **2021_04** was current from 17 Nov 2021 until 22 Feb 2022, and **2022_01** became current only from 23 Feb 2022 — so 22 of February's 28 days belong to 2021_04's window and only 6 to 2022_01's. We built our era-correct comparator (below) at

2022_01 (566,996 total entries; human subset by `OX=9606`, 20,376 proteins) as a **plausible candidate release**, not a confirmed one — the entry-count match to 2022_01's published size does not discriminate between the two candidates, since total Swiss-Prot moved by only ≈0.19% across the entire gap between them. We did not build 2021_04 and do not assert a number for it. Every headline is additionally re-derived against the 2022_01 build, which is the estimand relevant to whether the overlap was detectable when the resource was built, whichever of the two candidate releases IEAtlas actually used. Both the modern and the 2022_01 numbers are reported.

The overlap measurement

A catalogued peptide sequence counts as canonical-compatible if it is an **exact substring** of at least one protein in *R*. Sequences are compared as unmodified amino-acid strings, uppercased, with inline PTM annotations stripped. Peptides are **deduplicated to unique sequences** before rates are computed; a rate is `unique canonical-compatible / unique scored`. The unit is a peptide **sequence**, never an atlas row. The identical procedure is applied to every catalogue we reprocess, so that comparison is internally consistent. Rates published by other groups under other pipelines are cited as context and **never** combined into a ratio with ours.

Length standardization. Because exact-substring probability is strongly length-dependent, rates are also reported after **direct standardization** to a common length distribution (IEAtlas's own), with per-length rates given unpooled. A standardized rate is not reported for a catalogue with fewer than 20 overlap events, where it would be falsely precise.

The library measurement

For each ncORF library, all distinct **9-mers** (a representative HLA-I ligand length) are enumerated across the **full library — no sampling** — and intersected with the set of all distinct 9-mers in *R*. Sampling is invalid here and the bias is large and upward: sampling 4,000 of nuORFdb's 229,251 ORFs gives 43.6%, 20,000 gives 40.7%, and the full library gives 34.1%, because a small sample contains fewer distinct ncORF-specific k-mers and so over-weights the canonical-shared ones. Whole-ORF containment *is* sampled (4,000 ORFs, seed 0); that statistic is an $O(n \cdot m)$ substring scan and is unbiased under random sampling, unlike a k-mer union.

Union caveat, and the union itself. For libraries *A*, *B* and canonical set *C*, the combined rate is $|(A \cup B) \cap C| \;/\; |A \cup B|$, which is **not** monotone in the addition of *B*. No bound on the combined library's ambiguity is claimed from nuORFdb alone. The union of the two sources we hold is measured directly (nuORFdb ∪ Translnc = 24.3%, *below* nuORFdb's 34.1%, restricting Translnc to its human-annotated entries — its distributed FASTA also carries 148,667 mouse-convention headers, whose inclusion is reported only as a diagnostic at 20.2%). RPFdb v2.0 is not measured and is **not approximated**: the live site serves only v3.0, and it distributes RibORF genomic *coordinates* rather than amino-acid sequences, so back-translation would require many free parameters and would yield *our* library rather than IEAtlas's. Its contribution is carried as a single free unknown *m*, giving a distribution-free cap of **53.5%** (human-only union; 47.6% under the whole-file diagnostic) on the three-source library and **no positive lower bound derivable from the sources we hold**.

A comparison we do not make. The library rate (distinct 9-mers of an undetected candidate space) and the catalogue rate (distinct peptides at native lengths, after search, FDR and deduplication) are **different objects with different units and denominators**. We report no difference, ratio or excess between them. The detection test is stated only as a within-catalogue contrast (breadth) and a **ratio of ratios** (ribosomal enrichment in the catalogue against the same enrichment in the library), both of which are immune to the unit mismatch.

The detection-bias test

IEAtlas records the cancer type of each observation (15 types). Breadth of detection (number of distinct cancer types per unique sequence) is compared between canonical-compatible and canonical-incompatible sequences, **stratified by peptide length**. The ribosomal-enrichment test compares the share of ribosomal-gene ORFs (`RPL*/RPS*/MRPL*/MRPS*`) between the two groups *in the catalogue*, against the same share computed over **distinct 9-mers in nuORFdb** (a 9-mer is ribosomal-associated iff it occurs in at least one ribosomal-gene ORF) — the baseline library-composition figure — against the composition prior to any detection. nuORFdb alone is not the full three-source library; the R3 body reports the sensitivity of this test to folding Translnc in as well. This is the same unit as §R3's library ambiguity rates; raw (ORF, 9-mer) occurrence pairs would double-count a 9-mer once per ORF containing it and are not unit-matched to the catalogue side, so we use the distinct-9-mer unit throughout. The reported effect is the **excess** of the former over the latter.

Pseudogene-contamination check. `RPL*/RPS*/MRPL*/MRPS*` symbol matching cannot itself distinguish a protein-coding ribosomal gene from its pseudogene (e.g. *RPS3AP12*, S1) — a class already established to be canonical-compatible for reasons unrelated to abundance. Each ribosomal-prefixed symbol observed in either the catalogue or nuORFdb is resolved to an NCBI GeneID (`Homo_sapiens.gene_info`) and checked against the same curated pseudogene→parent registry (NCBI `gene_group`) used for S1, splitting the class into true protein-coding ribosomal genes, ribosomal-named pseudogenes, and symbols the registry does not resolve. Recomputing the catalogue-side and library-side ratios separately within each split (nuORFdb-only baseline): true genes give catalogue RR 3.13× / library RR 0.90× = excess **3.47×**; pseudogenes give catalogue RR 1.35× / library RR 0.62× = excess **2.16×**. Both splits recombine to the pooled 2.51×/0.69×/3.66× reported above. The excess is not an artifact of pseudogene contamination of the "ribosomal" label.

The normal-tissue consequence

IEAtlas's cancer and normal exports are compared as unique sequences. The canonical-incompatible sequences of the same catalogue are a **within-resource comparator**, not a control: they do not control abundance, detectability, HLA coverage or study composition. Inference uses a **gene-clustered bootstrap** — source-gene clusters resampled with replacement, B = 2,000, seed 20260713 — of the **length-standardized** risk ratio and, as the primary detection-opportunity-adjusted analysis, the risk ratio directly standardized on peptide length × cancer-detection-breadth bins (1, 2, 3, 4, 5+ distinct cancer-type labels). Only strata containing both groups contribute; their pooled distribution supplies the common standard. Exact unbinned breadth is a sensitivity analysis. A **tissue-clustered bootstrap** is reported separately under the same resampling scheme. A peptide carrying more than one gene or tissue label (§1) is assigned to a single cluster deterministically — the lexicographically minimal label in its set — so no peptide is counted in more than one cluster and no resampling draw can double-count it. A two-proportion z-test is **not** valid here (the observations are clustered) and is not used.

External processed-table crosswalk. PRIDE's project and file APIs were queried for PXD038782. All 37 files categorized `SEARCH` whose names contained `_benign_` were streamed and parsed; inline parenthetical modifications were removed and sequences were uppercased and deduplicated before exact intersection with the IEAtlas cancer catalogue. Tissue label and preparation class were parsed from the deposited filename (`W6-32`, HLA-I preparation; `Tue39L243`, HLA-II preparation). The 37 tables contained 219,760 valid peptide rows, all labelled `Found By = PEAKS DB`. Counts are therefore reported as processed-table recurrence. No raw-spectrum re-search, spectrum-level USI verification, donor-allele match, or source-locus inference was performed.

Class strata

ORF-class strata are **mutually exclusive** (`pseudogene-only`, `non-pseudogene-only`, `both`), because a peptide may carry several ORF labels; a pseudogene / non-pseudogene split is *not* a partition and double-counts the 546 sequences carrying both.

Class-specific FDR identifiability

Derivation of Θ_N(q, f) and its sharpness **given q and f** in the Supplement. The underlying phenomenon is not new (Woo et al. 2014 measured it empirically), and neither is the bound's general form (a Manski-style partial-identification bound, Manski 1989); the closed-form instantiation for class-conditional FDR is absent from Woo et al. 2014. Stated here because IEAtlas reports q but not f.

ImmunoVerse recurrence and raw-lineage pilot

We used the 2025-07-07 versions of ImmunoVerse Supplementary Tables S7 (`ORF_antigen`) and S3 (study, raw file, biological label and reported HLA genotype). Peptide sequences and biological labels were joined exactly. HLA strings were normalized only by uppercasing and removing `HLA-` and `*`. Sequence recurrence was counted first across Table S7 sample/condition labels and then bounded across Table S3's published source-study identifiers. For a label mapping to several studies, the minimum and maximum compatible study counts were retained; no study was selected by convention. The primary peptide–HLA sensitivity included published strong and weak binding predictions. The conservative ratio divided the minimum compatible sequence-study recurrence by the maximum compatible recurrence of the best predicted peptide–HLA.

Peptides were ranked by decreasing sample/condition recurrence, then lexicographically by sequence, with a prespecified top 1,000 and a tie-complete sensitivity analysis.

For raw linkage, we exhaustively matched the historical BLCA and DLBC Table S7 sequences to the authors' current consolidated `msmsScans` tables, then used Table S3 to map raw filenames to biological labels and reported genotypes. Exact sequence recovery, label-set recovery, PSM-count agreement, current `Identified` status and `Reverse` flags were retained separately. The scan tables post-date the historical supplement, so current acceptance decisions were **not** treated as reproducing the historical consolidation policy. An empty intersection of complete reported HLA genotypes rules out one common allomorph across those contexts; it does not identify the presenting allele in any context, resolve the source locus, establish independent-patient recurrence, or estimate catalogue-level FDR.

Reproducibility

Every headline number in this manuscript is regenerated from the committed artifacts by `manuscript/verify_manuscript.py`, **which fails the build on drift** and additionally enforces the paper's required prior-art citations and a fixed list of banned phrasings. The analysis code is guarded by `src/darkproteome/scoring_conformance.py`. All primary Methods quotations were verified in the fetched full text (EuropePMC / NCBI E-utilities); the fetch scripts and document hashes are in the repository. Every entry in the References list below was independently re-verified against a publisher, PubMed, or DOI-resolver record, including each citation's title against its source.

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Alphabetical by first author surname. Verification methodology (which entries were checked against fetched full-text XML vs. publisher/PubMed records) is in `data/SOURCES.md`.

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Supplement

Companion to the main manuscript. Three items sit here rather than in the main text, for three different reasons: **S1** is presented as an **exploratory supplementary analysis**, not a headline result; **S2** is an elementary derivation of a phenomenon Woo et al. 2014 established empirically — the closed form itself is not established prior art (checked directly against Woo et al. 2014) — included because the main text depends on its exact form; **S3** is a frozen, version-specific worked example from a separate atlas, included to test whether sequence recurrence survives aggregation into a peptide–HLA recurrence claim.

S1. Where it can be interrogated, the ambiguity is structured by homology

This quantifies a known phenomenon. That a processed pseudogene shares sequence with its parent gene, and that the resulting peptides are not attributable to one locus, is the classical shared-peptide / protein-inference problem (Nesvizhskii & Aebersold). We measure it; we do not discover it.

Why this is in the Supplement. Two methodological concerns apply to any symbol-heuristic parent mapping and a naive permutation null: parentage **derived from the gene symbol** (strip a trailing `P` + digits) is not authoritative, and a permutation null that treats parents as freely exchangeable ignores **gene-family structure**. Both are addressed below — with an authoritative annotation and a family-respecting null — and the result survives both. It stays here anyway, for two reasons the reader should weigh: the testable set is **133 peptides** out of 174,465, and the authoritative mapping is **not independent** of the symbol heuristic it replaces (see the caveat below). It is a mechanistic vignette that corroborates the main result; it is not itself a headline measurement.

The authoritative parent mapping

Parentage is now taken from **NCBI Gene `gene_group`** — the curated *"Related pseudogene"* relation (build 2026-07-13; 13,437 human pseudogenes with a curated parent; no multi-parent cases), with symbols resolved through `Homo_sapiens.gene_info`, HGNC (`hgnc_complete_set`, build 2026-07-10) including previous and alias symbols, and GENCODE v26 as a clone-name→ENSG bridge. (pseudogene.org's psiDR and PseudoPipe were downloaded and **not used**: they are built on GENCODE v7/hg19 (2012) and Ensembl 90 (2017) respectively, too stale to adjudicate current symbols.)

Coverage, stated plainly. Of the 62 distinct pseudogene symbols carrying a canonical-substring peptide, **36 resolve to a curated parent (58.1%)**. The remaining 26 split into two different reasons, not one: **25** are clone-style identifiers (`AC005262.1`, `AL158050.1`, …) absent from **every** registry, and **1** (`GNG5P2`) resolves cleanly to an NCBI GeneID but that GeneID carries no curated `gene_group` parent relation at all — a real gene symbol with no annotated parent, distinct from the clone-style cases, which have no symbol-to-GeneID resolution to begin with. 36 + 1 + 25 = 62. Among symbols that have a real gene name (37 = 36 + 1), coverage is **97.3%**.

Of the 36 resolved-parent symbols, only 35 carry a testable peptide. One symbol's curated parent is `H2BC8`; Swiss-Prot has no reviewed entry under that symbol — not because the protein is missing, but because UniProt collapses several histone-H2B genes that encode an identical protein into one reviewed entry filed under a different symbol (`H2BC4`), which gene-ID-level matching cannot see through. That symbol is therefore excluded from the denominator (the conservative call — it is not scored as a miss), leaving **35** testable genes, matching the "33/35" gene-level row below.

The caveat that matters, and it is not a small one. HGNC *names* a pseudogene after the parent it descends from. The curated relation is therefore **not an independent source of truth** about parentage — it is substantially the same information the symbol heuristic was reading, curated. High agreement between the two is **expected, and is not corroboration**. What the curated mapping actually buys is a *versioned, adjudicated* relation with correct handling of renames and family cases — which is the substantive improvement over the symbol heuristic — and nothing more. The **test** itself is untainted:

whether a peptide occurs inside the parent's protein is a pure sequence question, decided independently of any naming.

The measurement

A processed pseudogene is a retro-copy of a parent gene, so a peptide encoded by a pseudogene ORF can be an exact substring of the **parent protein**. Of the pseudogene-labelled IEAtlas peptides that are canonical substrings, **133 are testable**:

	authoritative	**(symbol heuristic)**
land in the pseudogene's **own curated parent**	**77 / 133 = 57.9%**	79 / 132 = 59.8%
at gene level, pseudogenes with ≥1 peptide in their own parent	**33 / 35**	33 / 34

Table S1. Pseudogene-to-parent hit rate under the authoritative NCBI Gene mapping versus the symbol-heuristic mapping it replaces.

How wrong was the heuristic? Where both mappings yield a parent, they **agree in 124 / 127 = 97.6%** of cases; it disagrees 3 times, and in 7 further cases the heuristic yields no parent where a curated one exists. **Seven peptide-level verdicts flip.** The headline moves from 59.8% to **57.9%** — a small, real correction: the symbol heuristic was directionally right, but not authoritative. We report both.

This head-to-head comparison population is **124 + 3 + 7 = 134 peptides, not 133**: it counts every peptide whose pseudogene *symbol* resolves to a curated parent, regardless of whether that parent has its own reviewed SwissProt entry. The **133**-peptide "testable" denominator used everywhere else in this section additionally requires the parent to have a SwissProt entry, which excludes exactly the one peptide noted above (`HIST1H2BPS2`, parent `H2BC8`) — so 134 = 133 + 1. That one peptide still has a well-defined curated-parent *symbol* to compare against the heuristic even though it cannot be scored for the hit-rate test itself.

The null — and one that turned out to be degenerate

Permuting parent labels freely ignores gene-family structure, which can make "matched the parent" trivially easy to satisfy by chance. We ran four nulls (10,000 permutations, seed 0) and report all of them, including the one that failed:

null	construction	result
A — naive / free	parents exchangeable across the whole pool (*the naive null*)	77 obs vs **4.4** null mean; p < 1e-4
B — HGNC gene-family	shuffle parents only within a curated family	restricted to its **5** permutable items: 5/5 (100.0%) obs vs **3.4** (67.8%) null mean; p = 0.20 (**not significant at this n**)
C — shared-9-mer component	shuffle only within a sequence-sharing component	**DEGENERATE** — see below
D — family-decoy swap	replace the true parent with a random **close paralog of that parent**	see below — *the null to read*

Table S2. Four family-respecting nulls for the pseudogene→parent hit rate; only Null D is non-degenerate.

Null B is not degenerate — it is small, and we report it rather than folding it in with C's true degeneracy. Of the 35 parent proteins, only 5 belong to a curated HGNC family with another parent in the pool at all; restricted to that permutable subset, all 5 (100%) land in their own parent against a null mean of 3.4 (67.8%). At n = 5 this cannot reach significance (p = 0.20) and we do not claim it does — it is reported for completeness, not as independent support. **Null C is the one that is genuinely degenerate**, and for a reason that is itself the answer to the objection: **the 35 parent proteins are pairwise 9-mer-disjoint**, so every homology stratum is a singleton and the "family-respecting" permutation reduces to the identity. There is no family structure *among the parents* for a shuffle to exploit. Reporting C's p = 1.0 as if it were evidence would be as dishonest as hiding it.

Null D is the family-respecting null that actually runs. It asks the sharp version of the objection: *could a merely homologous protein have been hit instead of the true parent?* Each parent is replaced by a random **close paralog of itself** — the hardest decoys available — under two pools:

decoy pool	true parent hit	random paralog hit	*p*
any shared 9-mer (permissive)	53 / 101 = **52.5%**	**7.1%**	< 1e-4
strong paralogs (≥10 shared 9-mers)	34 / 65 = **52.3%**	**16.6%**	< 1e-4

Table S3. Null D, the family-decoy swap, under two paralog-decoy pools.

The hit is **parent-specific, not family-generic** — it survives even against the parent's genuine close homologs. Without any null at all: **58 / 77 = 75.3%** of parent-hits are compatible with the parent and with **no** close paralog. The remaining 24.7% are not, and we say so; family structure is real and quantified rather than denied.

The objection was right for the reason it gave: curated mapping corrected one heuristic false match. The old symbol rule treated two symbols as the same gene if they shared a ≥3-character prefix with a numeric remainder — so it judged `ZNF720` and `ZNF135` the same gene. `ZNF720P1`'s peptide `KSFSHSSSL` occurs in *ZNF135*, *ZNF256* and *ZNF483*, and **not** in its true curated parent *KRABD5* (of which `ZNF720` is merely a previous symbol). The heuristic scored a parent hit by zinc-finger string collision; the curated relation deletes it. **Aggregate cost of that failure mode: one hit.**

What it does and does not license

It does show that canonical-sequence compatibility is *concentrated in the curated parent* — the ambiguity sits exactly where descent predicts, and is parent-specific rather than family-generic (null D). The pseudogene class label is corroborated by sequence structure rather than being arbitrary.

It does not resolve provenance. `DEVAFRKF` is encoded by both *RPS3AP12* and *RPS3A*; MS identifies the sequence, not the locus. A parent hit is not even a *unique* assignment — many parent hits also match further canonical genes. This is **more** ambiguity, not less.

It is not a surprise, and we do not dress it up as one. A processed pseudogene *is* a degenerate copy of its parent; that its peptides match the parent's protein is the mechanism working as expected. The value is explanatory — it says *why* a large part of the pseudogene class is source-ambiguous — not evidentiary.

The 56 non-parent matches (42.1% of 133) are an unresolved residual. Out-of-frame retro-copy ORFs and incidental short-peptide matches are both plausible. **We report this and do not explain it.**

Across other classes we report only descriptive heterogeneity in canonical overlap. The retro-copy mechanism is claimed for **processed pseudogenes only** — the one class where the label is corroborated by sequence. Low overlap in lncRNA-ORF (0.5%) and altORF (0.2%) classes argues against one specific failure mode (wholesale pseudogene-like contamination of those classes) and nothing more; it does not validate those annotations.

S2. Set-identification of a class-specific FDR (elementary derivation; the underlying phenomenon is Woo et al. 2014)

What kind of quantity *q* and *FDP_N* are, stated before the derivation, because "FDR" invites conflating two different things. Everything below is an algebraic identity about **realized proportions in one fixed, already-accepted set of identifications**: it partitions an actual count of false discoveries into two actual per-class counts, and needs no assumption about repeated draws, exchangeability, or *f* being non-random. For one realized accepted set there is only one set — the identity below holds by the arithmetic of a weighted average of two group proportions, not by a probabilistic argument about FDR as an expectation over a hypothetical ensemble of experiments.

Separately: q, as a pipeline reports it, is itself typically a target–decoy **estimate** of that realized proportion (e.g. a decoy-count-based ratio at a chosen threshold), not a directly observed ground truth, and decoy-based FDR estimation carries its own assumptions and estimation noise. $\Theta_N(q, f)$ below is sharp *given* q and f taken as exact; if q is itself estimated with error, that error propagates into Θ_N and is not additionally bounded by anything here.

Let a target–decoy procedure accept a set of identifications at a reported pooled false-discovery rate q. Partition the accepted set into a non-canonical class N and its complement, and let f be the fraction of accepted identifications in class N.

Write FDP_N for the **realized false-discovery proportion within class *N*** — the actual (unknown) fraction of accepted class-N identifications that are false, in this one accepted set — and FDP_C for that in the complement. We use *FDP*, not *FDR*, throughout this derivation: *FDR* is an expectation over a hypothetical ensemble of experiments, and q itself is typically a target–decoy **estimate** of a realized proportion, not that proportion directly. The identity below is arithmetic on realized counts and needs neither name to hold, but calling the unknowns *FDP* keeps it from reading as a claim about the expectation. The pooled realized proportion is the f-weighted mixture:

$$p = f \cdot FDP_N + (1 - f) \cdot FDP_C$$

with the only constraints being that both class-specific proportions are in [0, 1]:

$$0 \le FDP_N \le 1, \quad 0 \le FDP_C \le 1$$

Solving for FDP_N and imposing those bounds on FDP_C gives

$$FDP_N = (p - (1 - f) \cdot FDP_C) / f$$

which is decreasing in FDP_C. Substituting the two extremes $FDP_C = 1$ and $FDP_C = 0$, and intersecting with [0, 1]:

$$\Theta_N(p, f) = [\max(0, (p - (1 - f)) / f), \min(1, p / f)]$$

Sharpness. Every point in Θ_N is attained by some admissible $FDP_C \in [0, 1]$, and no point outside it is: the endpoints correspond exactly to $FDP_C = 1$ and $FDP_C = 0$. So **given *p* and *f* alone, Θ_N is the sharp identified set for the realized class-*N* false-discovery proportion** — the data-generating process is not pinned down more tightly by that information. Substituting a *reported* q for p evaluates Θ_N at an estimate of the realized proportion, not at the proportion itself; if q has estimation error, that error is not additionally bounded by anything here (see below).

What this does *not* say. Θ_N is sharp *with respect to p and f*, evaluated in practice at q as a stand-in for p. It is **not** the claim that no further information could narrow it. Calibrated class-specific posterior error probabilities, entrapment measurements, or a validated mixture model would each add information and could tighten the interval. The per-class accepted decoy count D_N is simply the **cheapest sufficient** such object, and one the pipeline already computes: with T_N and D_N and the stated threshold, unit and convention, the selected class-specific target–decoy estimate becomes **reconstructible**. D_N does not identify the true class-specific false-discovery *proportion*, and we do not claim it does.

Why it bites here. IEAtlas reports q = 0.05 as a PSM-level target–decoy threshold, but publishes no PSM-level canonical/non-canonical accept counts at all — not even the class-summed, cross-tissue 245,870 epitope total is at the right unit for this (main text, R4). f is therefore unknown by any route we have found, and Θ_N(0.05, f) is **unconstrained** — for small f the upper endpoint min(1, q/f) reaches 1. The interval cannot be evaluated at all from what the resource publishes. That is the reporting gap, and it is closed by one small table.

Worked illustration. At q = 0.05:

f (non-canonical share of accepted IDs)	Θ_N(0.05, *f*)
0.50	[0.00, 0.10]
0.20	[0.00, 0.25]
0.05	[0.00, 1.00]
0.01	[0.00, 1.00]

Table S4. Worked values of the set-identified class-specific FDR interval Θ_N(0.05, f) at four plausible class fractions.

The scarcer the class, the less a pooled threshold says about it — which is the entire point of Woo et al. 2014, restated here only because the resource under audit does not report the quantity that would resolve it.

S3. Recurrence and scan-level provenance in a frozen ImmunoVerse pilot

Scope and estimand

This is a **worked external example**, not a pan-atlas prevalence estimate and not an audit of the live 2026 ImmunoVerse portal. The catalogue and sample map are frozen to the 2025-07-07 preprint supplements: Table S7 (`ORF_antigen`) and Table S3 (source study, raw file, biological label and reported HLA genotype). Table S7 contains 17,741 source rows representing **7,770 distinct peptides**. The current raw-result release post-dates those tables and is used only to test evidence recoverability in two bounded cancer subsets.

Recurrence is reported at four distinct units: sample/condition label, published source-study identifier, peptide sequence, and predicted or genotype-compatible peptide–HLA. A source-study label is not assumed to be an independent patient, cohort or experiment. Table S7's HLA entries are binding predictions, not direct allele assignments.

Catalogue-to-study lineage

Table S3 contains **1,771 rows**, **1,679 distinct raw-file names**, **498 biological labels**, and **47 source-study identifiers**. All **451 / 451** biological labels used by Table S7, and all **465 / 465** cancer–label pairs, joined exactly to Table S3. Six used labels map to more than one source study; their study recurrences were bounded rather than resolved by convention. Two labels carry conflicting reported genotype variants. After formatting normalization, 33,707 Table S7 prediction entries match every reported genotype for their label, 41 match some but not every reported genotype variant, and none mismatch all reported genotypes.

Quantity, top 1,000 sequence-recurrent peptides	Result
median sample/condition-label recurrence	**11**
median compatible source-study recurrence	**7–7**
median conservative label/source-study ratio	**1.667**
label recurrence exceeds the study upper bound	**97.2%**
median conservative sequence-study/best-predicted-pHLA-study ratio	**1.250**
conservative pHLA ratio > 1	**72.2%**
conservative pHLA ratio ≥ 2	**16.3%**

Table S5. Source-study normalization under the primary strong-or-weak predicted-binder rule. The conservative pHLA ratio divides the minimum compatible sequence-study recurrence by the maximum compatible study recurrence of the best predicted pHLA.

The top-1,000 boundary cuts through a 240-peptide tie at seven labels. In the tie-complete set (n = 1,184), the median conservative pHLA ratio remains 1.250 and 72.1% remain above one. In the 732-peptide exact-study-mapping subset, the median sequence recurrence is 6 studies, the best predicted pHLA recurs in 5, the median ratio is 1.333, and 81.6% exceed one. Under strong binders alone, 913 of the top 1,000 have a predicted pHLA; the median conservative ratio is 1.333 and 73.8% exceed one.

Current raw-result recovery: a positive control

The current release publishes MaxQuant, rescoring and consolidated scan outputs separately from the catalogue. In BLCA, the consolidated table contains 106,026 scan rows. Every historical Table S7 sequence and count was recoverable:

BLCA recovery check	Result
exact sequence recovery	**16 / 16 peptides**
historical total `n_psm` / current exact-sequence rows	**18 / 18**
exact biological-label-set recovery	**16 / 16 peptides**
exact per-peptide PSM-count agreement	**16 / 16 peptides**
at least one current `Identified = +` scan	**15 / 16 peptides**
at least one current `Reverse = +` scan	**1 / 16 peptides**

Table S6. BLCA positive-control recovery. Current decision flags are reported as current state, not assumed to reproduce the historical acceptance policy.

This complete recovery shows why ImmunoVerse is a useful positive implementation example: the catalogue itself projects away the evidence joins, but the separately released raw-result chain makes them reconstructible.

DLBC: recurrence of a sequence is not recurrence of one pHLA

The current DLBC consolidated table contains **51,446 scan rows**. Of 73 historical DLBC Table S7 peptides, **72 / 73** were recovered by exact sequence and exact biological-label set. Historical `n_psm` summed to **194**, compared with **199** current exact-sequence scan rows; 68 / 73 peptides had exact per-peptide count agreement, 64 / 73 had at least one current `Identified = +` scan, and 11 / 73 had at least one current `Reverse = +` scan. These differences are decision/version provenance, not evidence that the historical catalogue was wrong.

All three multi-label sequences were recovered in every catalogue-listed label. Their complete reported genotype intersections and their common predicted-binder intersections were empty:

peptide	Table S7 labels	historical `n_psm`	current exact-sequence rows	current identified rows by label	common reported HLA
`AEGPDHHSL`	DOHH2, SUDHL4	5	5	DOHH2: 4; SUDHL4: 1	**none**
`VPHTRPVSL`	DOHH2, HBL1, SUDHL4	24	24	DOHH2: 6; HBL1: 7; SUDHL4: 7	**none**
`LASPHSPIL`	DOHH2, HBL1, SUDHL4	13	15	DOHH2: 0; HBL1: 0; SUDHL4: 0	**none**

Table S7. Raw-linked recurrent DLBC sequences. `AEGPDHHSL` and `VPHTRPVSL` each have a current accepted scan in every listed cell line, but the cell lines share no reported HLA allele. Conditional on those reported genotypes, one fixed peptide–HLA allomorph cannot explain recurrence across all contexts. `LASPHSPIL` is retained as a decision-version example: exact scans remain, but none carries the current consolidated acceptance flag.

Limits

This pilot does not establish independent-patient recurrence, the presenting allele for an individual scan, source-locus identity, malignant-cell specificity, T-cell recognition, safety, efficacy, or a catalogue-level FDR. It does establish the narrower estimand correction: a peptide sequence can recur across contexts in which one common pHLA target is impossible under the published genotypes. The separately published scan evidence makes this correction auditable; it should be propagated into the catalogue rather than left for every downstream user to reconstruct.